

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	19 July 2026
Team ID	
Project Name	Nutrition Assistant
Team Members	Kulasri Durga Deepika Bora Polimera Girija(Team lead)
Maximum Marks	4 Marks

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement / Description
FR-1	User Registration	Register through a validated form; prevent duplicate email accounts; securely hash passwords.
FR-2	User Authentication	Login using email/password; issue JWT token; restore session; logout and clear local authentication.
FR-3	Profile Management	View and update name, profession, age, gender, height, weight, goal and activity level.
FR-4	Meal Tracking	Create, read, update and delete meals with type, food, quantity, unit, calories, protein, carbohydrates, fats, notes and date.
FR-5	Water Tracking	Add quick/custom water amounts, view daily total and delete entries.
FR-6	BMI Management	Calculate BMI, classify category, save history, retrieve latest record and delete records.
FR-7	Dashboard Summary	Display profile, today's meal totals, water progress, latest BMI, recent meals and recommendations.
FR-8	Weekly Analytics	Aggregate seven-day calories, water, meals and macronutrients and display charts and tables.
FR-9	Personalized Recommendations	Generate rule-based suggestions using profession, health goal, meals, protein, calories, water and BMI.
FR-10	Authorization & Data Ownership	Protect APIs with JWT and ensure users can access only their own records.

Non-functional Requirements

NFR No.	Non-functional Requirement	Description
NFR-1	Usability	The interface shall be responsive, readable and easy to navigate with clear forms, validation

		messages, loading states and empty states.
NFR-2	Security	Passwords shall be hashed using bcryptjs; protected routes shall require JWT; secrets shall be stored in environment variables; CORS shall allow only approved frontend origins.
NFR-3	Reliability	API operations shall validate data, return meaningful status codes and handle errors without exposing sensitive information.
NFR-4	Performance	Dashboard queries shall use parallel database requests where appropriate; indexes shall support user/date retrieval; charts shall use aggregated API responses.
NFR-5	Availability	The local application shall remain available while MongoDB, backend and frontend services are running; cloud deployment can improve public availability.
NFR-6	Scalability	The three-layer architecture, modular routes/controllers/models and MongoDB collections shall support future features and additional users.
NFR-7	Maintainability	Frontend components and backend modules shall follow a consistent folder structure and reusable code patterns.